

From UAS to Satellite: Fractional Snow Cover Mapping of the Zugspitzplateau

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EARTH RESEARCH OBSERVATION

Introduction

Fractional snow cover (FSC) estimation describes the proportion of snow within a given pixel and can provide a more accurate representation of snow cover extent than binary classifications. FSC is an important metric to capture changing snow dynamics and improve hydrological modelling, as it enables more realistic representations of snowmelt patterns and runoff generations. This study investigates to what extent machine learning models trained on data obtained with an uncrewed aerial system (UAS) can estimate FSC of the Zugspitzplateau at a resolution of 10m, using Sentinel-2 data and a digital elevation model (DEM).

Methods

We collected four UAS datasets (*Tab. 1*) of a 51ha large study area on the Zugspitzplateau (*Fig. 1*) between September 2024 and June 2025. The data was obtained using a DJI Matrice 300 RTK. We created RGB orthomosaics at a spatial resolution of 2.5cm with a Zenmuse L1 sensor. Additionally, we obtained multispectral (RGB, red edge, near-infrared, and thermal) data with a MicaSense Altum sensor at a spatial resolution of 3.5cm for some of the observations. For each UAS observation, we obtained Sentinel-2 scenes with a maximum offset of four days, avoiding major snowfall and thaw events.

Sensor	Date	Cloud free Sentinel-2 Scene (*Date*)
RGB	2024-09-19	✓ (*2024-09-20*)
RGB	2024-10-25	✓ (*2024-10-24*)
Multispectral + RGB	2025-02-25	✓ (*2025-02-21*)
Multispectral + RGB	2025-06-04	✓ (*2025-06-06*) (2 scenes same date)

Table 1: *UAS acquisitions and corresponding Sentinel-2 Scenes*

The UAS data was used to create binary snow masks of the study area (*Fig. 2*), which are needed to calculate the FSC at lower resolutions. Snow classification used a pseudo-NDSI (Green - NIR / Green + NIR) augmented by thermal data. Cold pixels received a higher score boost to shift cold areas towards snow and suppressing warm surfaces not covered with snow, with per-scene binary thresholds derived via Otsu's method [1]. The scenes for which only RGB data was available were classified with a Random Forest Classifier (RFC). All binary classifications were aggregated to a resolution of 10m by calculating the fraction of snow-covered ground.

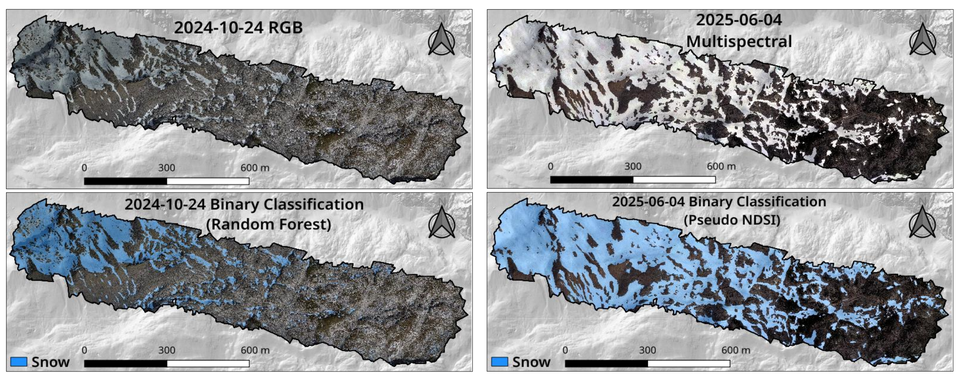


Figure 2: *Binary snow masks of two UAS scenes using the RFC (left) and pseudo-NDSI-based (right) approaches.*

We trained a Random Forest Regressor (RFR) and a Support Vector Regressor (SVR) to perform FSC mapping of the study area (*Fig. 3*). The input data consisted of ten Sentinel-2 bands (B2: blue, B3: green, B4: red, B5-7: red edge, B8: NIR, B8A: narrow NIR, B11: SWIR 1, B12: SWIR 2), three spectral indices (NDSI, NDVI, NDWI) and a snow-free DEM derived from LiDAR UAS acquisition on 20.08.2025. The first three scenes were used for training, while the June 2025 scene served as a validation dataset due to its highly variable snow conditions.

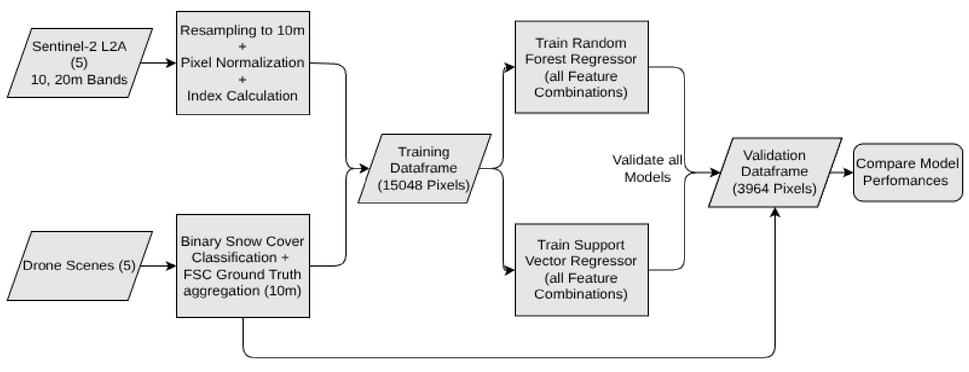


Figure 3: *Processing workflow to produce the fractional snow cover maps.*

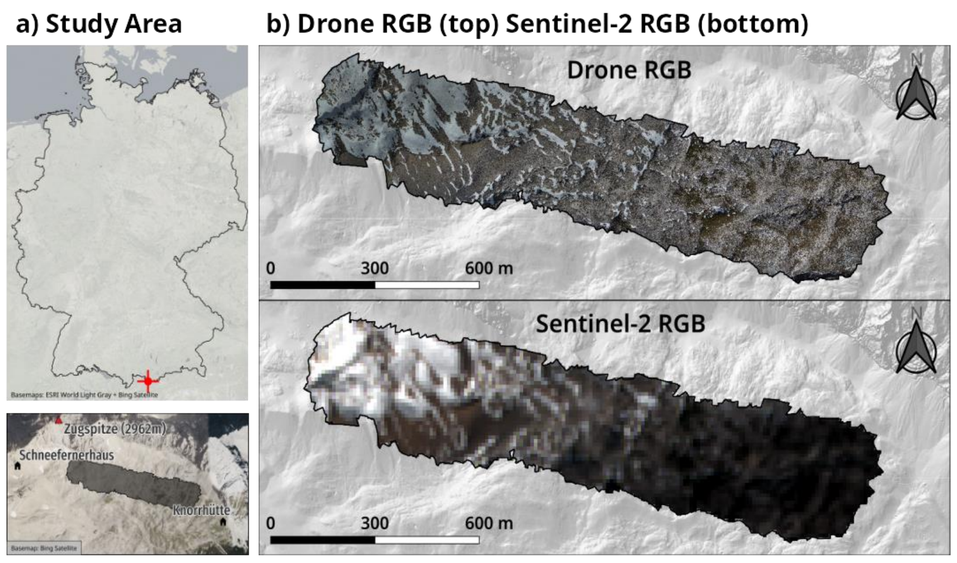


Figure 1: a) *Location of the study area,* ***b)*** *orthomosaics of the study area for the October 2024 acquisition.*

Results

We evaluated the accuracy of the snow masks using a validation set of 50 polygons per scene and found that the binary snow masking approach reliably distinguished between bare ground and snow. The RFC-based model reached a Kappa score of >0.96, while the pseudo-NDSI-based classifications had a Kappa score of >0.99.

For the FSC estimation, the RFR and SVR models showed good performances across different feature combinations. The RFR model performance was very stable, while the SVR model performance highly varied depending on the number of features. Both regressors heavily relied on spectral features and rarely used the DEM. The best results were obtained with an SVR model using the blue (B2), green (B3), and SWIR (B12) bands. This model had an R^2 value of 0.77 and an RMSE of 0.2. The resulting FSC map and associated ground truth FSC map derived from the binary snow classification can be seen in *Fig. 4*.

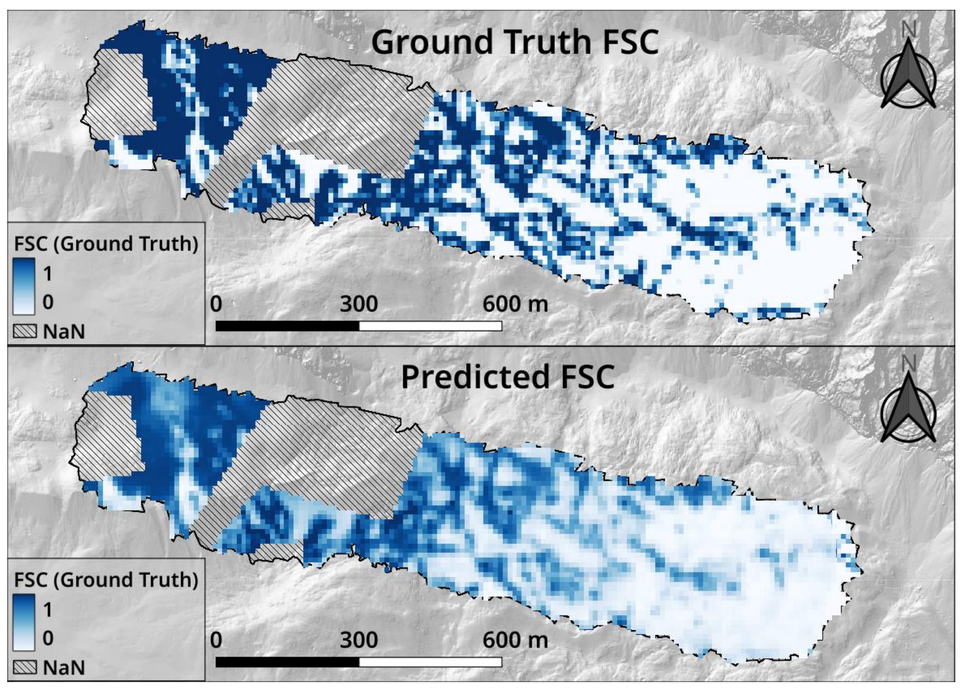


Figure 4: *FSC maps of the best performing model (SVR with features: B2, B3, B12). The ground truth FSC map uses UAS data acquired on 04.06.2025, while the predicted FSC map uses Sentinel-2 data acquired on 06.06.2025. In the western part of the study area, clouds were masked out and assigned as NaN.*

Conclusion

Our results indicate that a reliable FSC map can be created using multispectral satellite data with an SVR model validated using high resolution UAS data. Current limitations include the prediction of extreme values (0, 1). This is visible as off-white and light blue colors in the predicted map compared to pure white and dark blue in the ground truth (*Fig. 4*). Additionally, small pixel shifts between UAS-based ground truth and Sentinel-2 data remain a challenge to overcome. Nevertheless, our results show the potential of combining UAS and satellite data in in monitoring changing snow dynamics in alpine regions.

References

[1] N. Otsu, "A Threshold Selection Method from Gray-Level Histograms," in *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 9, no. 1, pp. 62-66, Jan. 1979, doi: 10.1109/TSMC.1979.4310076.